

Homework1

January 30, 2025

1 1.

```
[1]: def fibonacci(n):  
      x = 0  
      y = 1  
  
      numbers = []  
  
      for i in range(n):  
          numbers.append(x)  
          z = y  
          y = x + z  
          x = z  
  
      return numbers  
  
fibonacci(20)
```

```
[1]: [0,  
      1,  
      1,  
      2,  
      3,  
      5,  
      8,  
      13,  
      21,  
      34,  
      55,  
      89,  
      144,  
      233,  
      377,  
      610,  
      987,  
      1597,  
      2584,
```

4181]

2 2.

```
[2]: import math

def IsPrime(n):
    b = True
    for i in range(2,int(math.sqrt(n))+1):
        if n % i == 0:
            b = False

    return b

print(IsPrime(4))
print(IsPrime(23))
```

False

True

```
[3]: def primes(n):
    primes = []
    for i in range(2,n+1):
        if IsPrime(i):
            primes.append(i)

    return primes

primes(100)
```

```
[3]: [2,
3,
5,
7,
11,
13,
17,
19,
23,
29,
31,
37,
41,
43,
47,
53,
59,
```

61,
67,
71,
73,
79,
83,
89,
97]

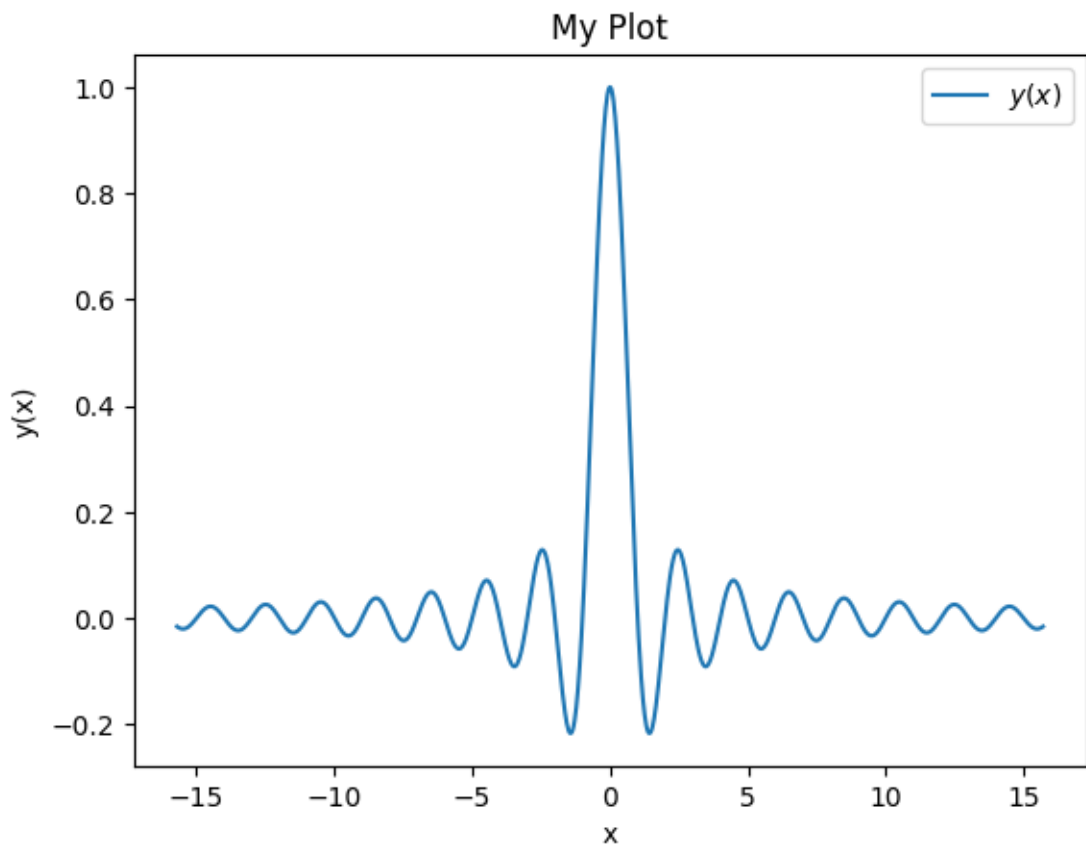
3 3.

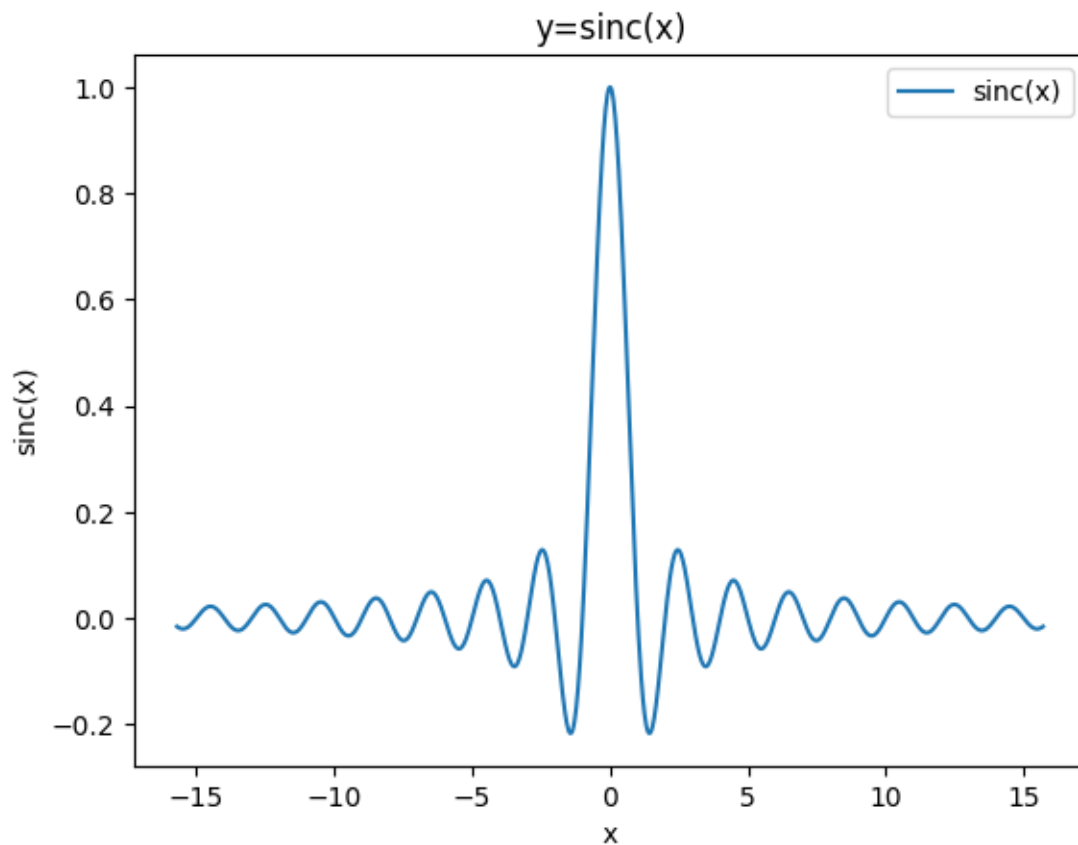
```
[4]: import numpy as np
import matplotlib.pyplot as plt

def MyPlot(x, y, xlabel='x', ylabel='y(x)', title='My Plot', label='$y(x)$'):
    plt.plot(x,y,label=label)
    plt.title(title)
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)
    plt.legend()
    plt.show()

x_vals = np.linspace(-5*np.pi, 5*np.pi, 1001)
y_vals = np.sinc(x_vals)

MyPlot(x_vals,y_vals)
MyPlot(x_vals,y_vals,ylabel='sinc(x)',title='y=sinc(x)',label='sinc(x)')
```





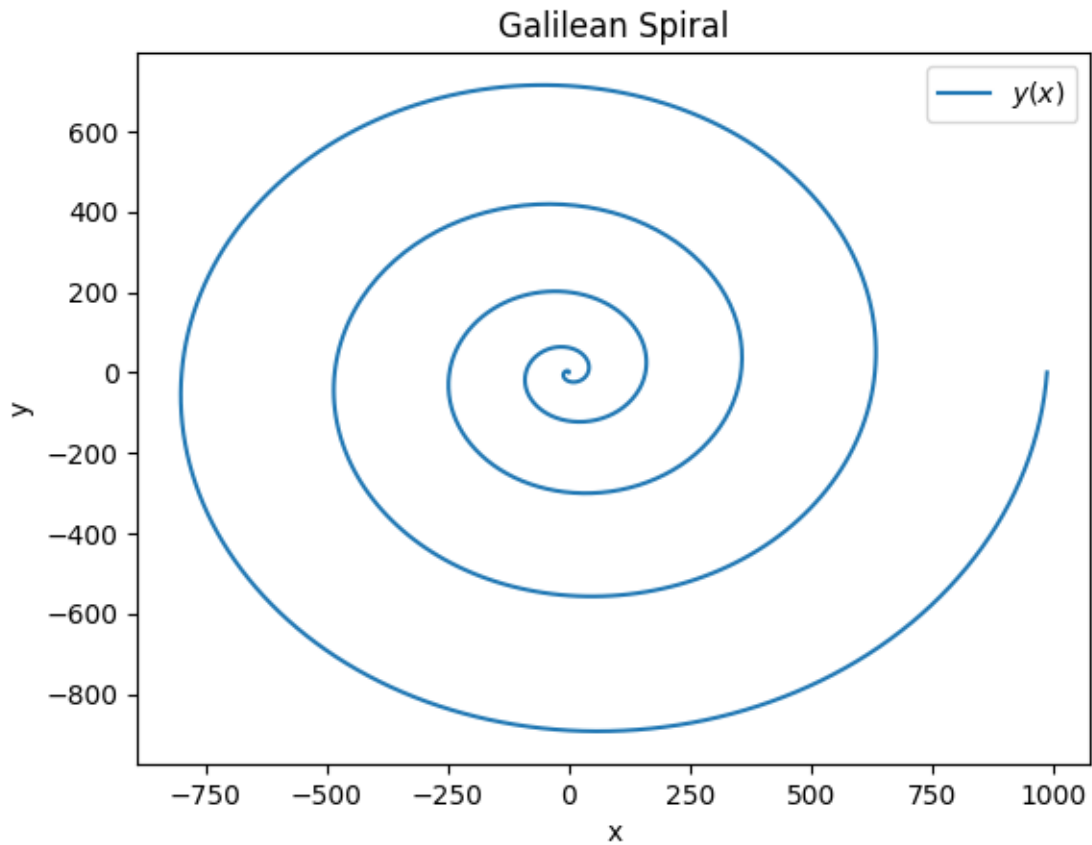
4 4.

4.1 a)

```
[5]: theta = np.linspace(0, 10*np.pi, 1001)
r = theta ** 2

x = r * np.cos(theta)
y = r * np.sin(theta)

MyPlot(x,y,title="Galilean Spiral",ylabel='y')
```



4.2 b)

```
[6]: theta = np.linspace(0, 24*np.pi, 5001)
r = np.exp(np.cos(theta)) - 2*np.cos(4*theta) + (np.sin(theta/12))**5

x = r * np.cos(theta)
y = r * np.sin(theta)

MyPlot(x,y,title="Fey's Function",ylabel="y")
```

